Institute of Engineering and Technology (IET)

Customer Churn Prediction & ROI-Backed Retention Playbook

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ABSTRACT

This project aims to predict customer churn in subscription-based businesses and design targeted retention strategies with measurable financial outcomes. Using machine learning techniques, the system will identify customers at risk of discontinuing services and provide actionable insights through interactive dashboards and a retention ROI calculator. The expected result is an integrated decision-support tool that not only predicts churn but also simulates the business impact of different retention interventions, helping companies maximize customer lifetime value.

INTRODUCTION

Background and Motivation:
Customer churn poses a critical challenge to subscription-based businesses such as telecom, SaaS, and streaming services. Losing customers directly reduces revenue, and retention is more cost-effective than acquiring new customers. Traditional churn detection approaches lack predictive accuracy and fail to quantify the financial benefits of retention strategies.

Relevance:
This project leverages machine learning, explainability tools, and business intelligence dashboards to provide both predictive and prescriptive insights. It connects data science with business decision-making, helping organizations design ROI-backed retention strategies.

Problem Statement:
Most companies struggle to identify churn risks early and evaluate the cost-effectiveness of retention campaigns. This project addresses the challenge by developing a predictive system with integrated ROI analysis to guide business decisions.

OBJECTIVES

- Predict customers who are likely to churn using machine learning models.
- Identify key drivers of churn using explainability techniques (SHAP).
- Provide actionable insights through dashboards and segment-level analysis.
- Develop a Streamlit-based ROI calculator to evaluate retention offers.
- Estimate Customer Lifetime Value (CLV) and enable targeted retention strategies.

SCOPE OF WORK

Included:

- Development of churn prediction models (Logistic Regression, Random Forest, XGBoost).
- Model explainability with SHAP values.
- Dashboard for churn insights (Power BI/Tableau).
- Retention ROI calculator (Streamlit).
- Customer segmentation and CLV estimation.

Excluded:

- Real-time data pipeline integration.
- Proprietary customer datasets (using publicly available Telco dataset only).
- Deployment beyond prototype scale.

METHODOLOGY

Step-by-Step Plan:

1. **Dataset Collection & Pre-processing:**
    - Use publicly available Telco Customer Churn Dataset.
    - Clean data, handle missing values, encode categorical variables.
2. **Exploratory Data Analysis (EDA):**
    - Analyze churn patterns by tenure, subscription plans, payment methods, etc.
    - Visualizations using Matplotlib/Seaborn.
3. **Model Development:**
    - Implement Logistic Regression, Random Forest, XGBoost.
    - Perform hyperparameter tuning and cross-validation.
4. **Model Evaluation:**
    - Use metrics such as Precision, Recall, F1-score, AUC.
    - Explainability through SHAP values to highlight churn drivers.
5. **Segmentation & CLV Analysis:**
    - Perform cohort analysis to estimate customer lifetime value.
6. **Dashboard & ROI Calculator Development:**
    - Create dashboards in Power BI/Tableau.
    - Build a Streamlit-based ROI calculator for financial impact analysis.
7. **Deployment & Documentation:**
    - Deploy the prototype using Streamlit Cloud/Local Server.
    - Prepare final project documentation and presentation.

Tools and Techniques:

- **ML & Programming:** Python, Pandas, Scikit-learn, XGBoost, TensorFlow/PyTorch.
- **Visualization:** Power BI, Tableau, Matplotlib, Seaborn.
- **Explainability:** SHAP.
- **Web App:** Streamlit.

EXPECTED OUTCOMES

- Early detection of customers likely to churn.
- Actionable insights for data-driven retention strategies.
- Quantifiable ROI from different interventions.
- Customer segmentation and CLV-based analysis.
- A fully functional prediction model, dashboard, and ROI calculator tool.

TIMELINE

Date Range	Task
Week 1	Requirement analysis, project setup, literature review, dataset collection & understanding
Week 2	Data preprocessing (cleaning, encoding, handling missing values), Exploratory Data Analysis (EDA)
Week 3	Feature engineering, initial model selection (Logistic Regression, Random Forest, XGBoost)
Week 4	Model training, hyperparameter tuning, cross-validation & evaluation using AUC, F1, Recall, Precision
Week 5	Model explainability with SHAP, identifying key churn drivers
Week 6	Customer segmentation (cohort analysis, CLV estimation), insights report preparation
Week 7	Dashboard development (Power BI/Tableau) and integration of churn prediction outputs
Week 8	Development of ROI calculator in Streamlit, simulation of retention strategies
Week 9	Deployment of Streamlit app & dashboards, internal testing & feedback
Week 10	Final refinements, documentation & project presentation